Machine Learning

Logistic Regression

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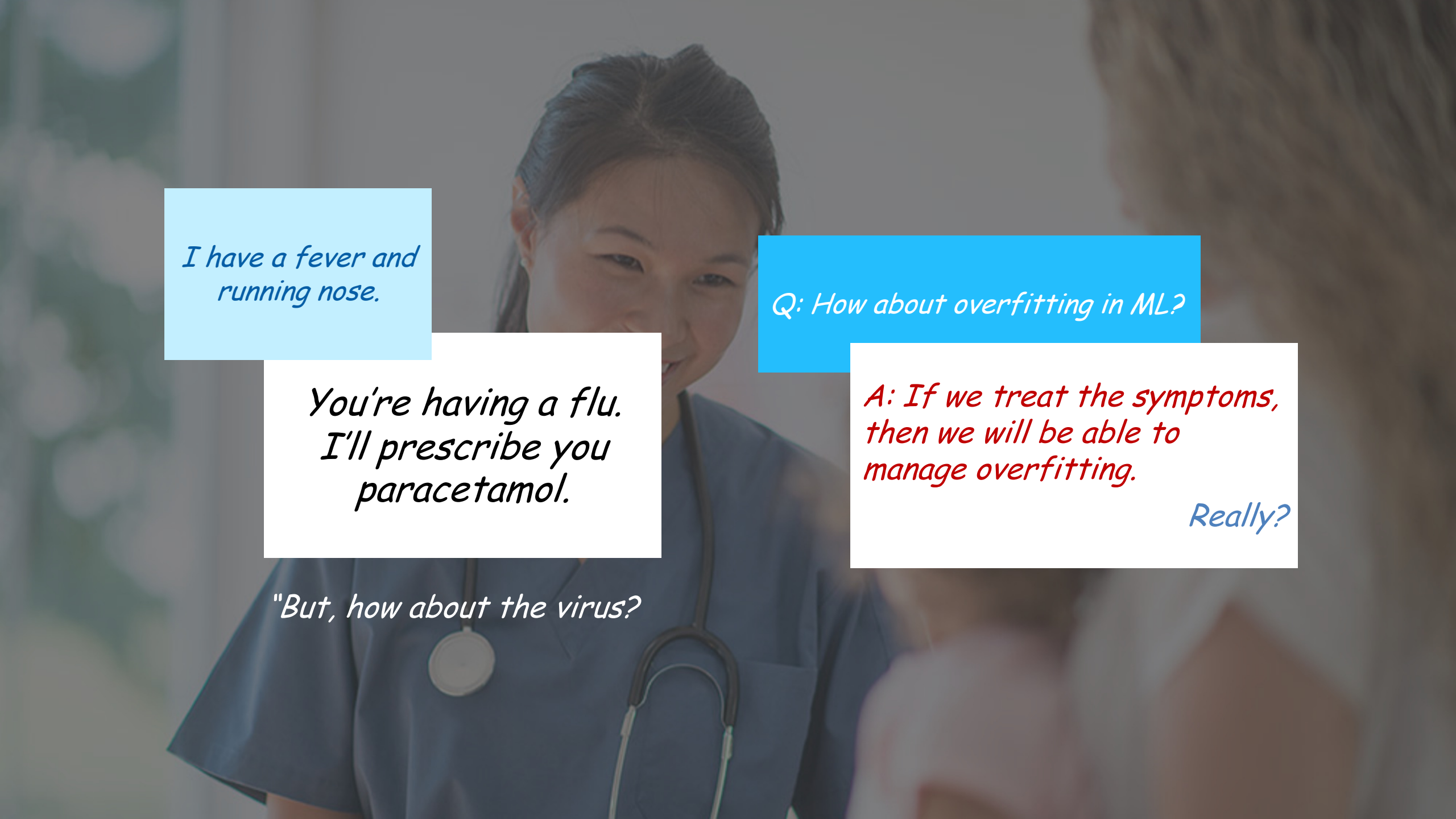
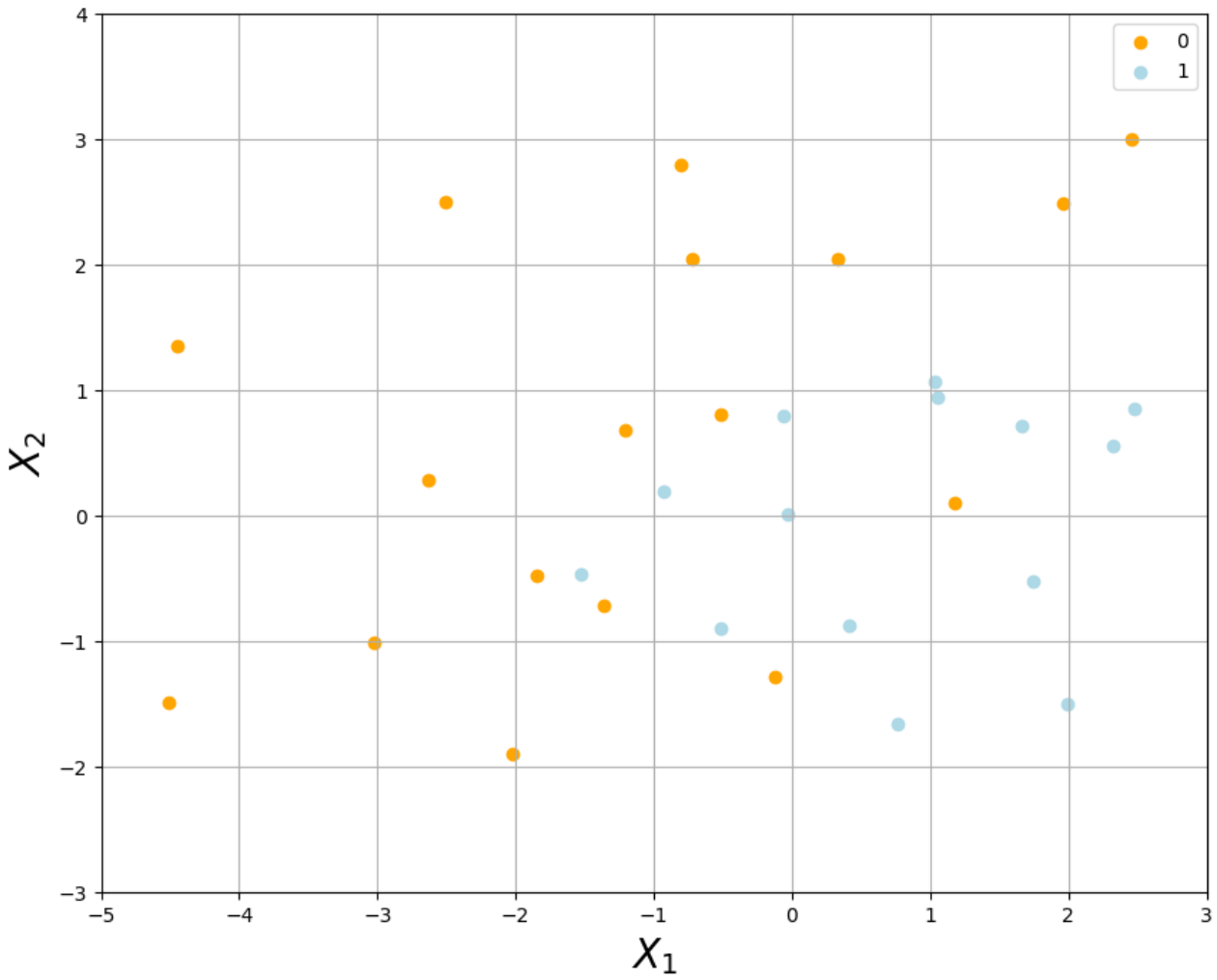
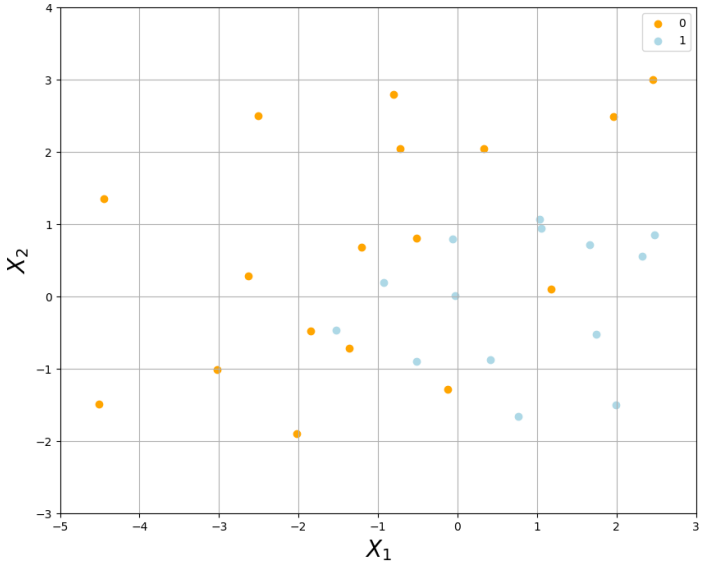

I have a fever and running nose.
You're having a flu. I'll prescribe you paracetamol.
"But, how about the virus?
Q: How about overfitting in ML?
A: If we treat the symptoms, then we will be able to manage overfitting.
Really?

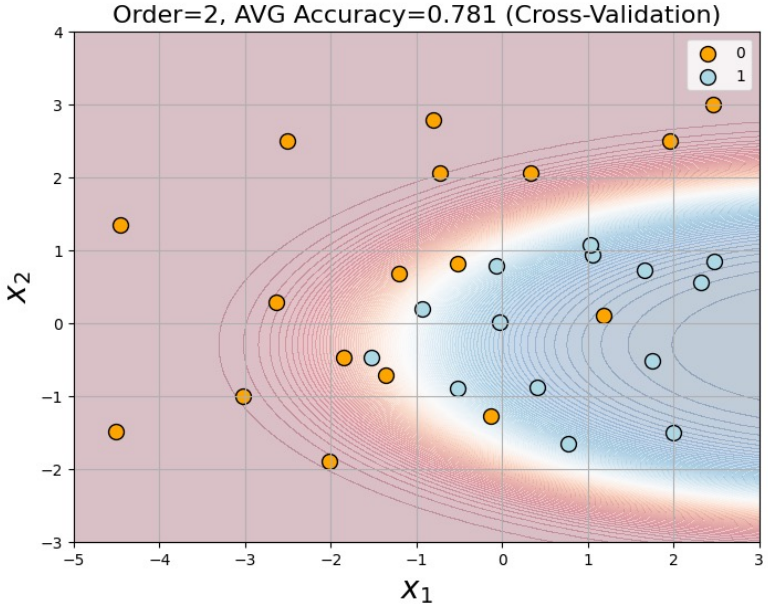
Example Dataset



Logistic Regression: Learn Decision Boundary (2nd Order)

Feature	Coefficient
1	1.717
x_1	1.385
x_2	-0.579
x_1^2	-0.167
x_2^2	0.978

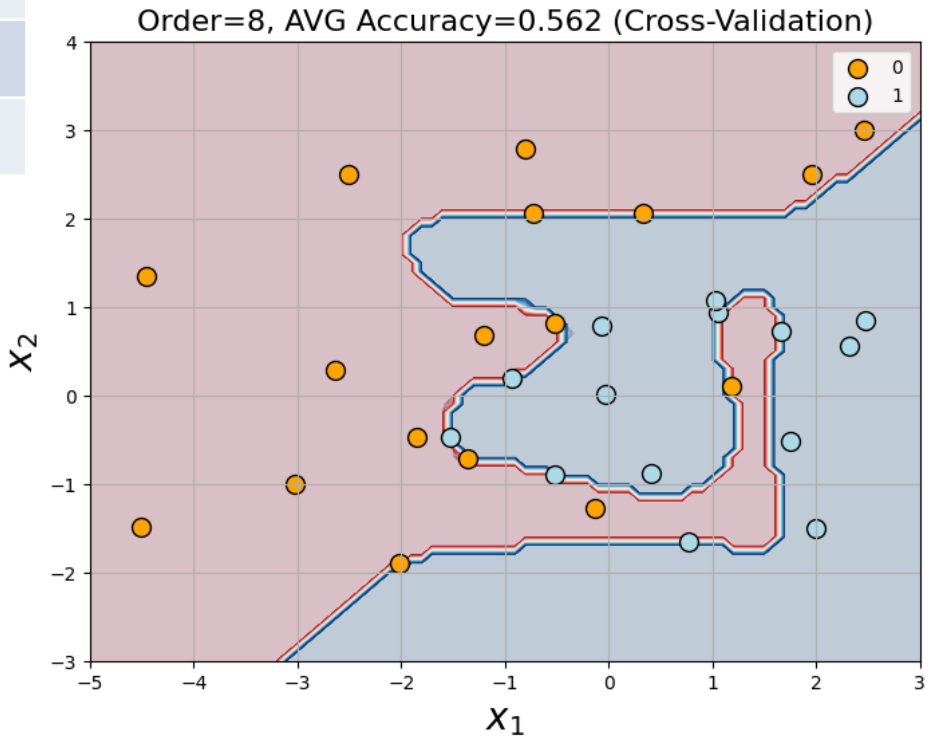




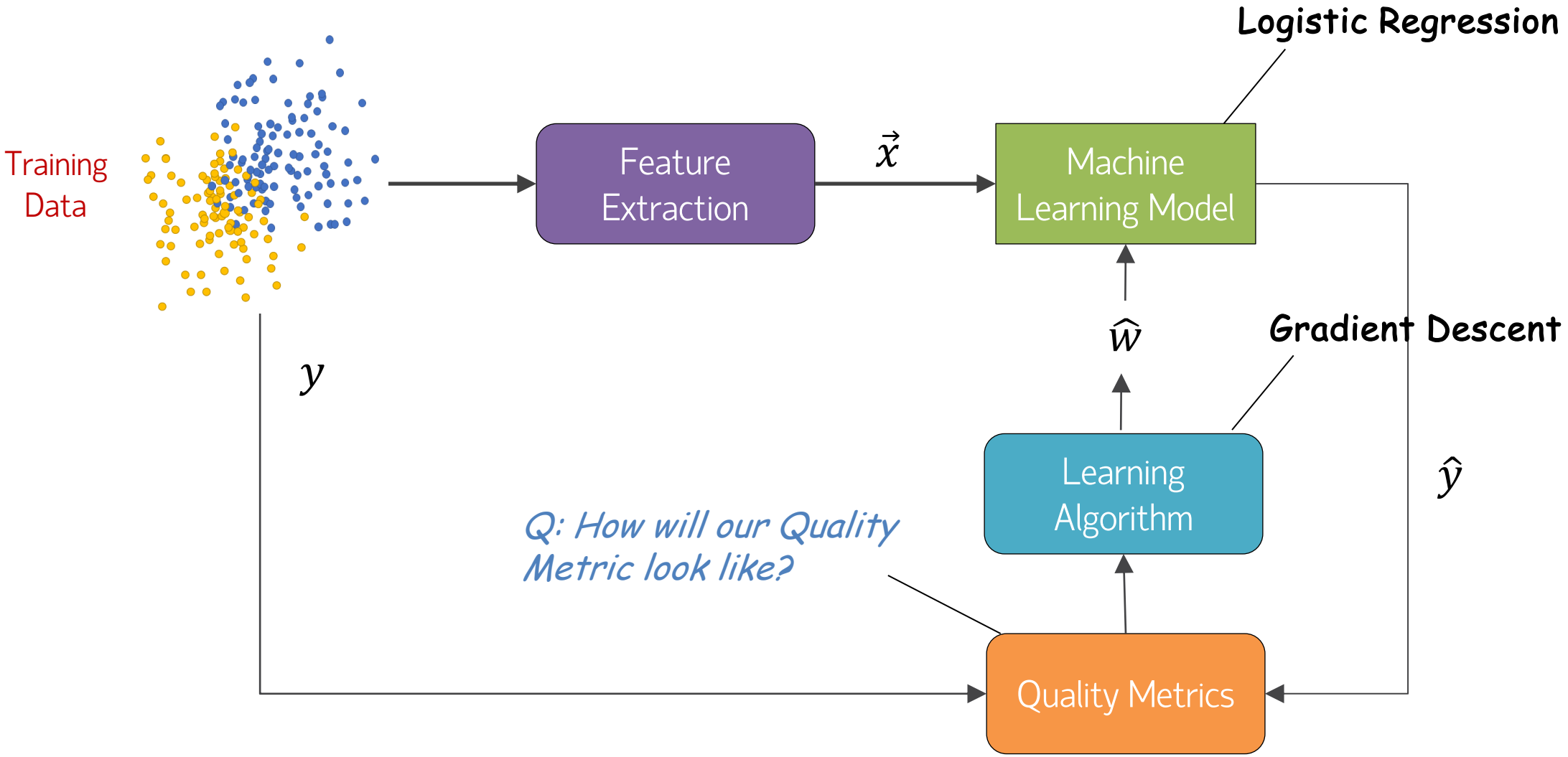
Logistic Regression: Learn Decision Boundary (8th Order)

Feature	Coefficient
1	587.984
x_1	495.169
x_2	-492.094
x_1^2	-76.356
x_2^2	-409.916
x_1^3	-53.310
x_2^3	286.153
x_1^4	-356.961
x_2^4	-120.300
x_1^5	-457.228
x_2^5	540.862

Feature	Coefficient
x_1^6	74.364
x_2^6	233.336
x_1^7	166.729
x_2^7	-167.182
x_1^8	17.001
x_2^8	-31.508



Workflow: Logistic Regression



Desired Total Cost

Want to Balance

Total Cost = Measure of Fit + Measure of Magnitude of Coefficients

$$-L(\vec{w})$$

Negative Log Likelihood

$$\|\vec{w}\|_2^2 = (w^{(1)})^2 + \cdots + (w^{(M)})^2$$

Resulting Objective

Minimising Total Cost Function:

$$\min_{\vec{w}} -L(\vec{w}) + \lambda \|\vec{w}\|_2^2$$

Tuning Parameter, i.e. Balance of Fit and Magnitude

Ridge Regression (aka. L2 Regularisation)

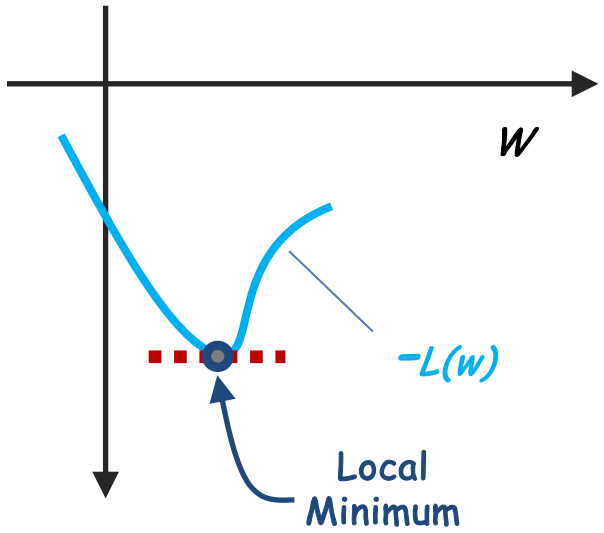
Gradient Vector

Total Cost Function:

$$J(\vec{w}) \quad = \quad -\sum_{i=1}^{N}[y_i \log P(1|\vec{x}_i,\vec{w}) + (1-y_i)\log(1-P(1|\vec{x}_i,\vec{w}))] + \lambda\|\vec{w}\|_2^2$$

Derivative of Total Cost Function:

$$\frac{\partial J(\vec{w})}{\partial w^{(j)}} \quad = \quad -\sum_{i=1}^{N}(y_i - P(1|\vec{x}_i,\vec{w})\,)x_i^{(j)} + 2\lambda\vec{w}$$



Gradient Vector:

$$\nabla_{\vec{w}} J(\vec{w}) \quad = \quad -X^T(Y - P(1|X,\vec{w})) + 2\lambda\vec{w}$$

Pseudocode for Logistic Regression with L_2 Regularisation

```
Function Logistic_Regress(new_point, data_points, labels, learning_rate, max_iterations, tolerance, lambda)
    Begin
        // Step 1: Add bias term (column of ones) to data_points
        Add a column of ones to data_points: X

        // Step 2: Initialize weights (theta) to zeros
        Set weights = vector of zeros with the same length as the number of columns in X

        // Step 3: Train the logistic regression model using gradient descent with L2 regularization
        For iteration = 1 to max_iterations do
            // Step 3.1: Compute predictions for all data points in a single vectorized operation
            z = X * weights
            predictions = 1 / (1 + exp(-z))

            // Step 3.2: Calculate the gradient with L2 regularization
            errors = labels - predictions
            gradients = (transpose(X) * errors / m) - (lambda * weights / m)

            // Step 3.3: Update weights
            weights = weights + learning_rate * gradients

            // Step 3.4: Check for convergence
            If the magnitude of all elements in gradients < tolerance then
                Break the loop

        // Step 4: Add bias term to new_point
        Add a 1 (bias term) to new_point: x  // x is of shape (n+1, 1)

        // Step 5: Predict the probability for new_point
        z_new = transpose(weights) * x
        probability = 1 / (1 + exp(-z_new))  // Sigmoid function

        Return probability
    End
```

Python Code Snippet

```
def Logistic_Regress(new_point, data_points, labels, learning_rate=0.01, max_iterations=1000, tolerance=1e-6,
lambda_=0.1):
    # Step 1: Add bias term (column of ones) to data_points
    X = np.hstack([np.ones((data_points.shape[0], 1)), data_points])

    # Step 2: Train the model using gradient descent with L2 regularization
    weights = gradient_descent_logistic(X, labels, learning_rate, max_iterations, tolerance, lambda_)

    # Step 3: Add bias term to new_point
    x = np.hstack([1, new_point])

    # Step 4: Predict the probability for new_point
    probability = expit(x @ weights)

    # Return the predicted probability
    return probability
```

Python Code Snippet (cont.)

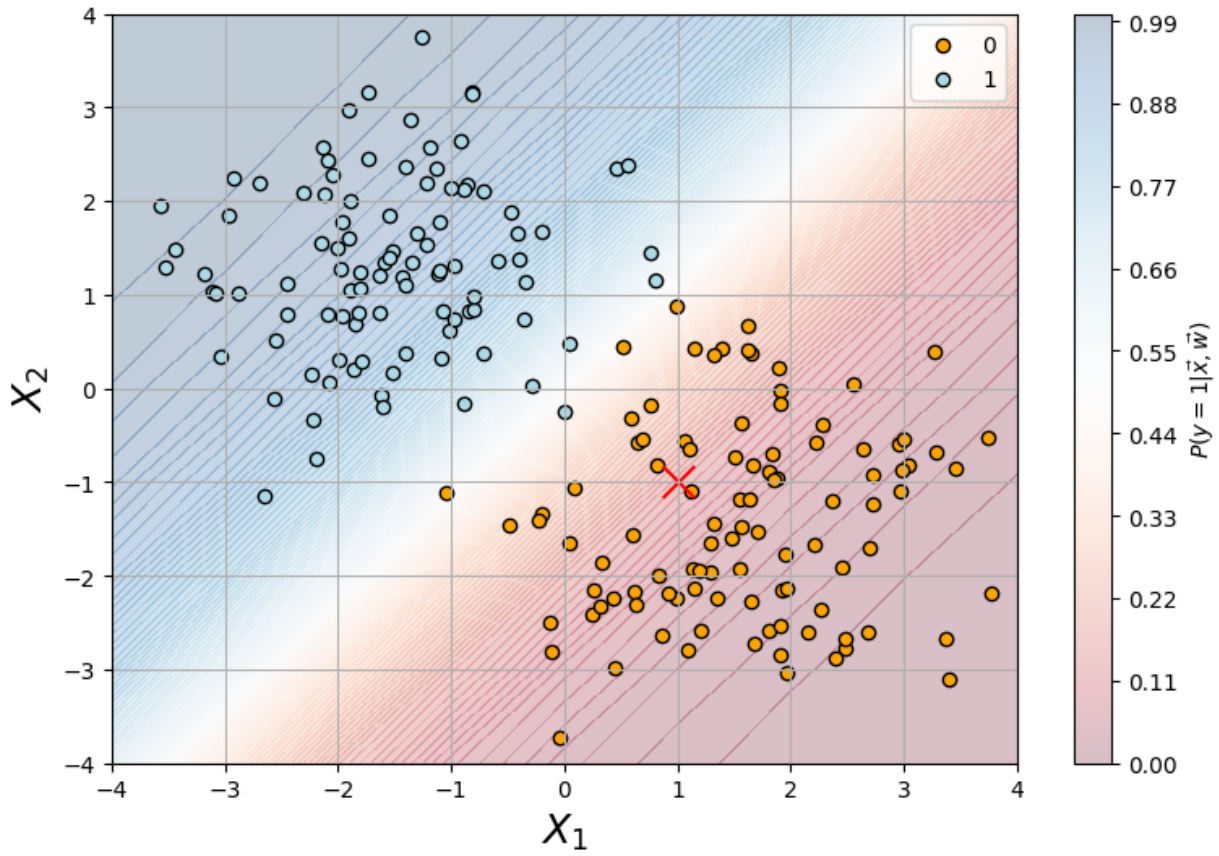
```
import numpy as np
from scipy.special import expit  # Optimized sigmoid function

def compute_gradient(X, y, weights, lambda_):
    predictions = expit(X @ weights)  # Predicted probabilities using sigmoid function
    errors = y - predictions           # Error between actual and predicted
    gradients = (X.T @ errors / X.shape[0]) - lambda_ * weights  # Regularized gradient

    return gradients

def gradient_descent_logistic(X, y, learning_rate=0.01, max_iterations=1000, tolerance=1e-6, lambda_=0.1):
    # Initialize weights
    weights = np.zeros((X.shape[1], 1))

    for iteration in range(max_iterations):
        # Compute gradient with L2 regularization
        gradients = compute_gradient(X, y, weights, lambda_)

        # Update weights
        weights += learning_rate * gradients

        # Check for convergence
        if np.linalg.norm(gradients) < tolerance:
            print(f"Converged after {iteration} iterations.")
            break

    return weights
```

Usage Example

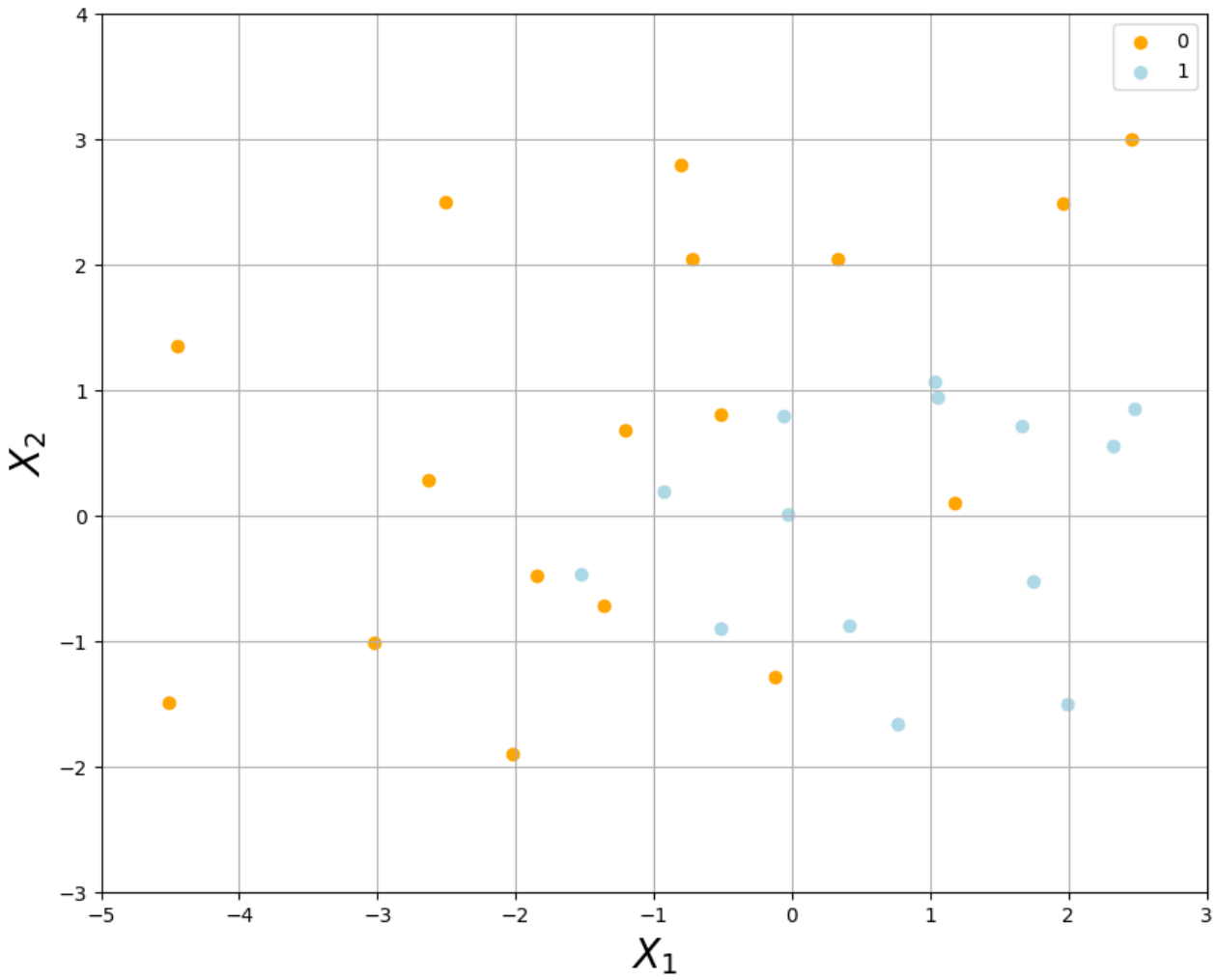
```
# Example usage
# Generate random data_points and labels for demonstration
# Set parameters for data generation
m, n = 100, 2 # Number of samples per class and number of features

np.random.seed(0)

# Generate data points for two classes
class_0 = np.hstack((1.5 + np.random.randn(m, 1), -1.5 + np.random.randn(m, 1)))
class_1 = np.hstack((-1.5 + np.random.randn(m, 1), 1.5 + np.random.randn(m, 1)))

# Combine the two classes into a single dataset
data_points = np.vstack((class_0, class_1))

# Generate labels: 0 for the first class, 1 for the second class
labels = np.vstack((np.zeros((m, 1)), np.ones((m, 1))))

# Define a new data point
new_point = np.array([1, -1])  # Example new point without bias term

# Predict the probability for new_point with L2 regularization
learning_rate = 0.1
max_iterations = 1000
tolerance = 1e-6
Lambda_ = 0.1

probability = Logistic_Regress(new_point, data_points, labels, learning_rate, max_iterations, tolerance, lambda_)

print(f"Predicted probability for new point: {probability}")
```

Usage Example

```
Converged after 416 iterations.
Predicted probability for new point: [0.1415866]
```

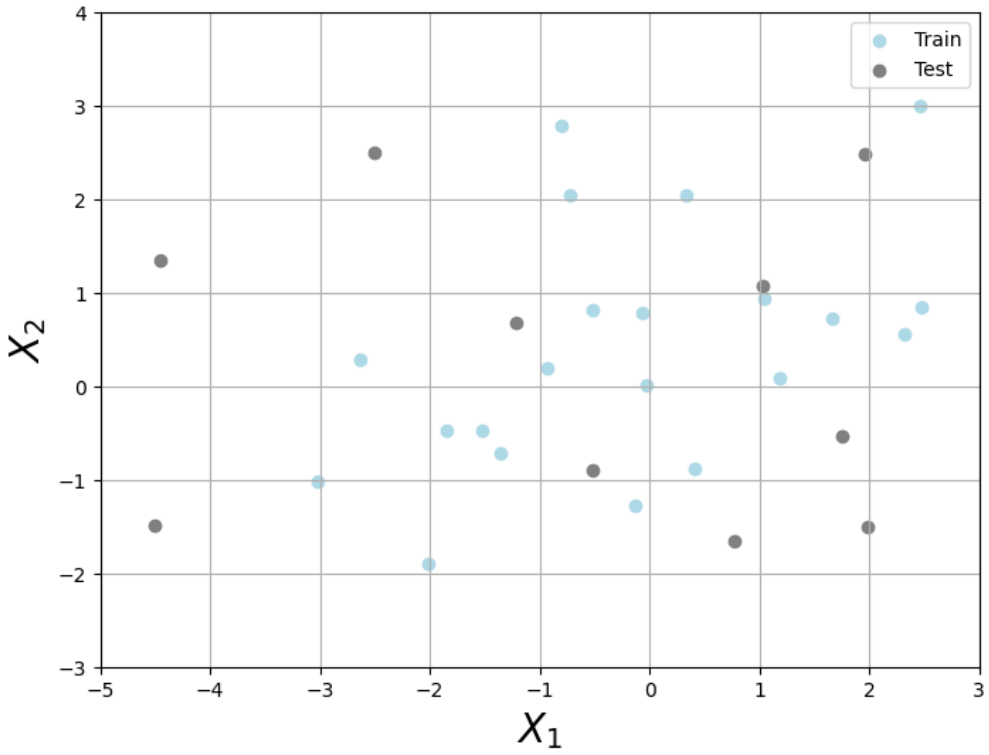


Example Dataset



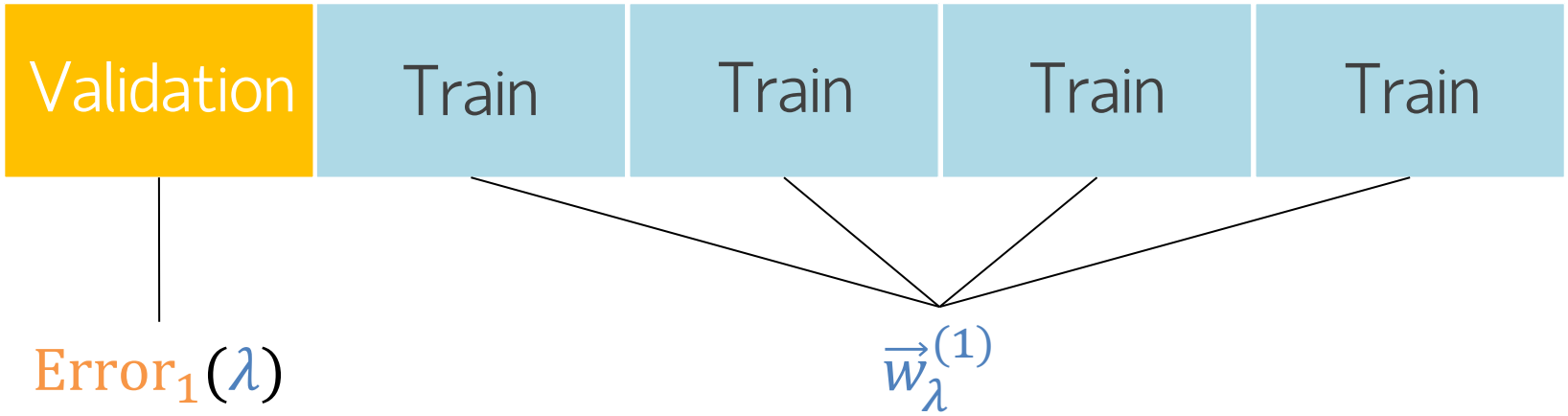
Train, Validation and Test Datasets

```
from sklearn.model_selection import train_test_split

# First, split the data into train (70%) and test (30%)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42, stratify=y)
```



- Train:Test is typicaly either 0.8:0.2 or 0.7:0.3.
- **Test** dataset is a proxy of unseen data, and it will only be used in the final evaluation.
- **Train** dataset is further divided into k folds (e.g., 5 or 10).For each fold, the model is trained on k-1 folds and validated on the remaining fold.
- We use **K-Fold Cross-Validation** to fine-tune or optimize the ML model. Here, we find a hyper-parameter λ based on the average performance across folds.

K-Fold Cross Validation



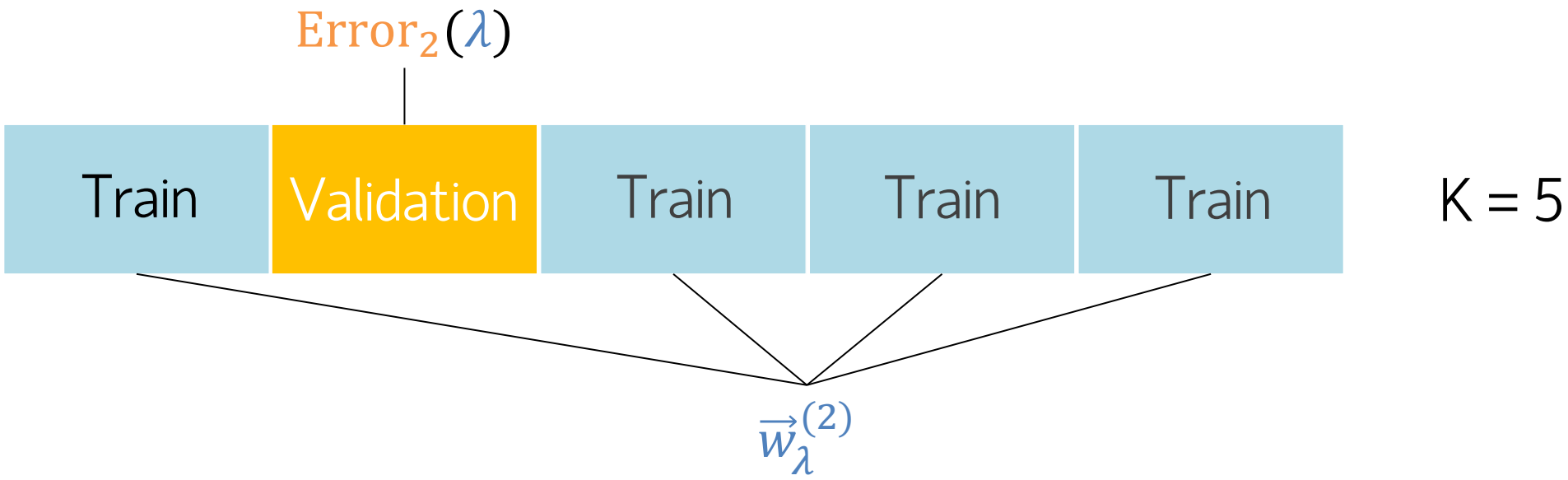
K = 5

For k = 1, ..., K

Step 1: Estimate $\vec{w}_\lambda^{(k)}$ on the training blocks.

Step 2: Compute error on Validation Block: $\text{Error}_k(\lambda)$.

K-Fold Cross Validation



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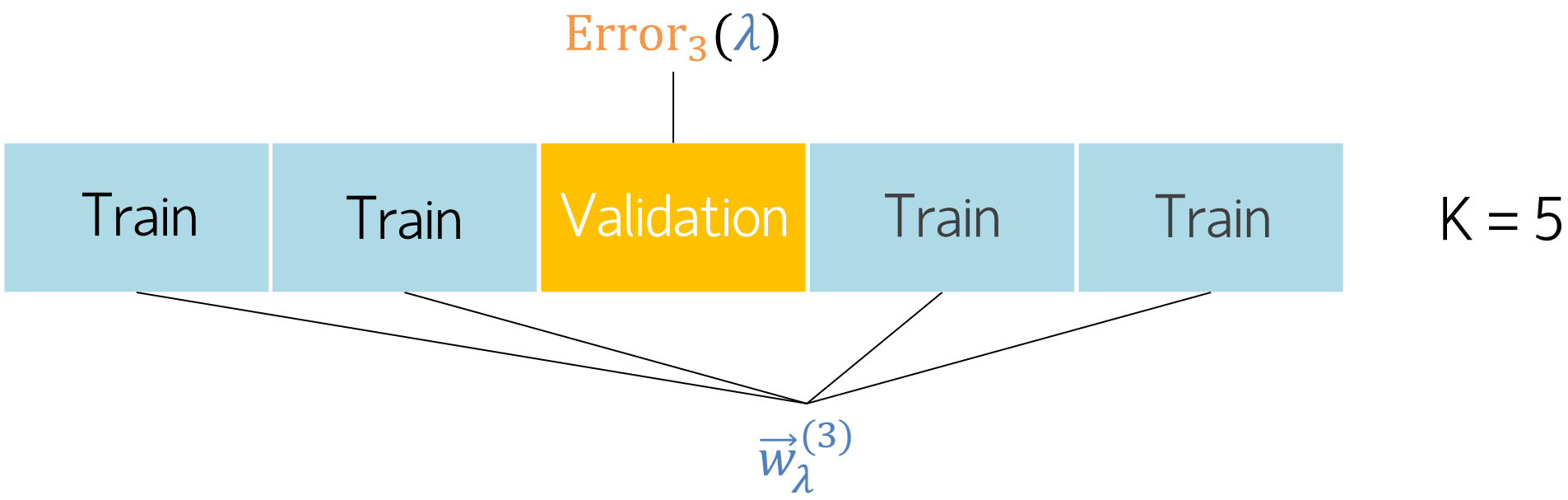

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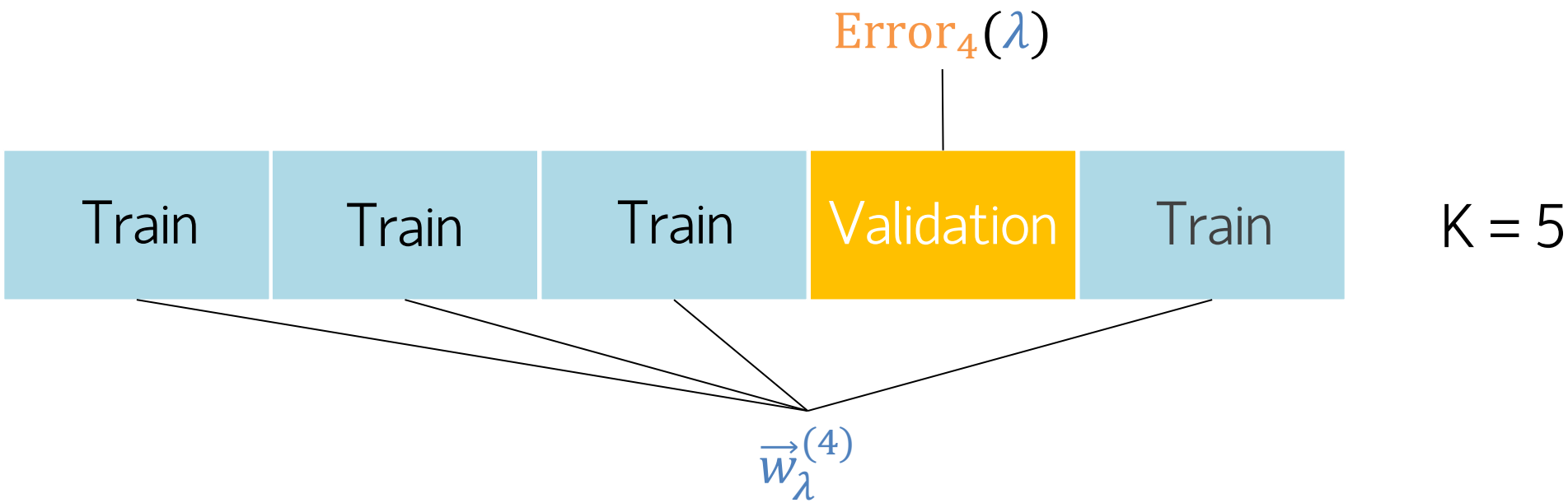
K-Fold Cross Validation



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For k = 1, ..., K

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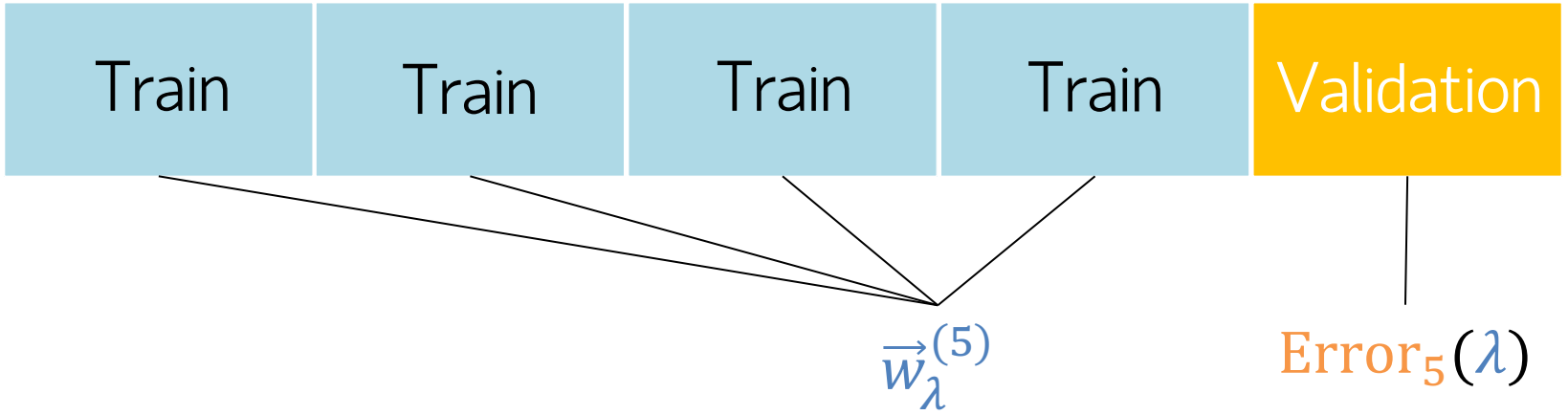
K-Fold Cross Validation



K = 5

For k = 1, …, K

Step 1: Estimate $\vec{w}_\lambda^{(k)}$ on the training blocks.

Step 2: Compute error on Validation Block: $\text{Error}_k(\lambda)$.

Compute Average Error: $\text{CV}(\lambda) = \frac{1}{K}\sum_{k=1}^{K} \text{Error}_k(\lambda)$.

scikit-learn: Logistic Regression with L_2 Regularisation

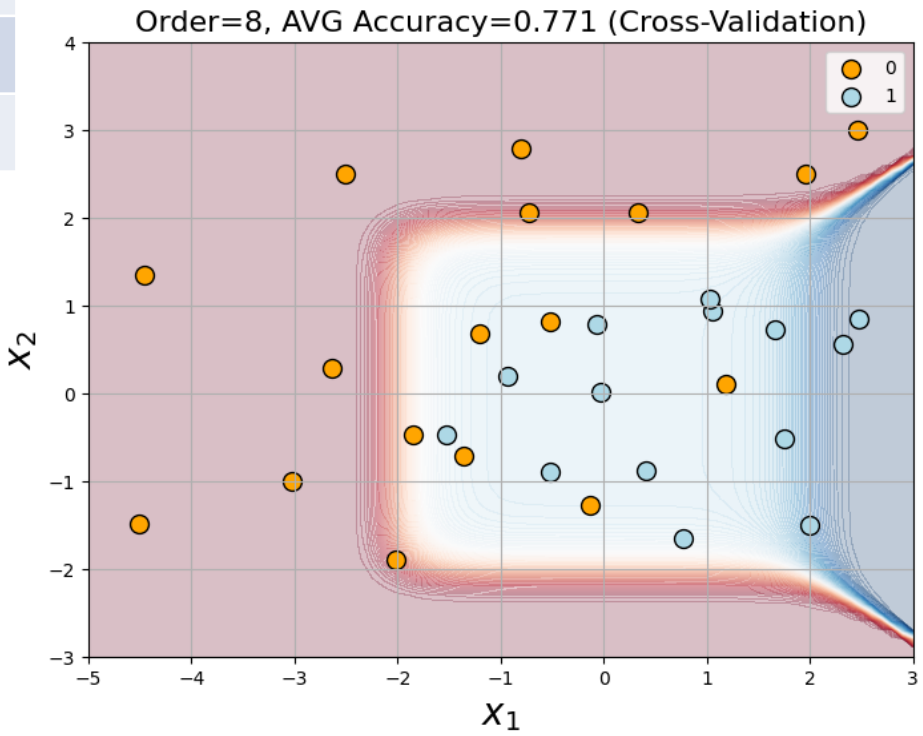
```
# Step 1: Standardize the features using only the training data
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Step 2: Create an instance of LogisticRegression
lambda = 10000
model = LogisticRegression(penalty='l2', C=1/lambda, random_state=204)

# Step 3: Define cross-validation strategy
k = 5
kf = KFold(n_splits=k, shuffle=True, random_state=42)

# Step 4: Perform cross-validation on the training data
cv_scores = cross_val_score(model, X_train, y_train, cv=kf, scoring='accuracy')
print("Mean Cross-Validation Accuracy: {:.2f}".format(np.mean(cv_scores)))
```

```
Mean Cross-Validation Accuracy: 0.77
```

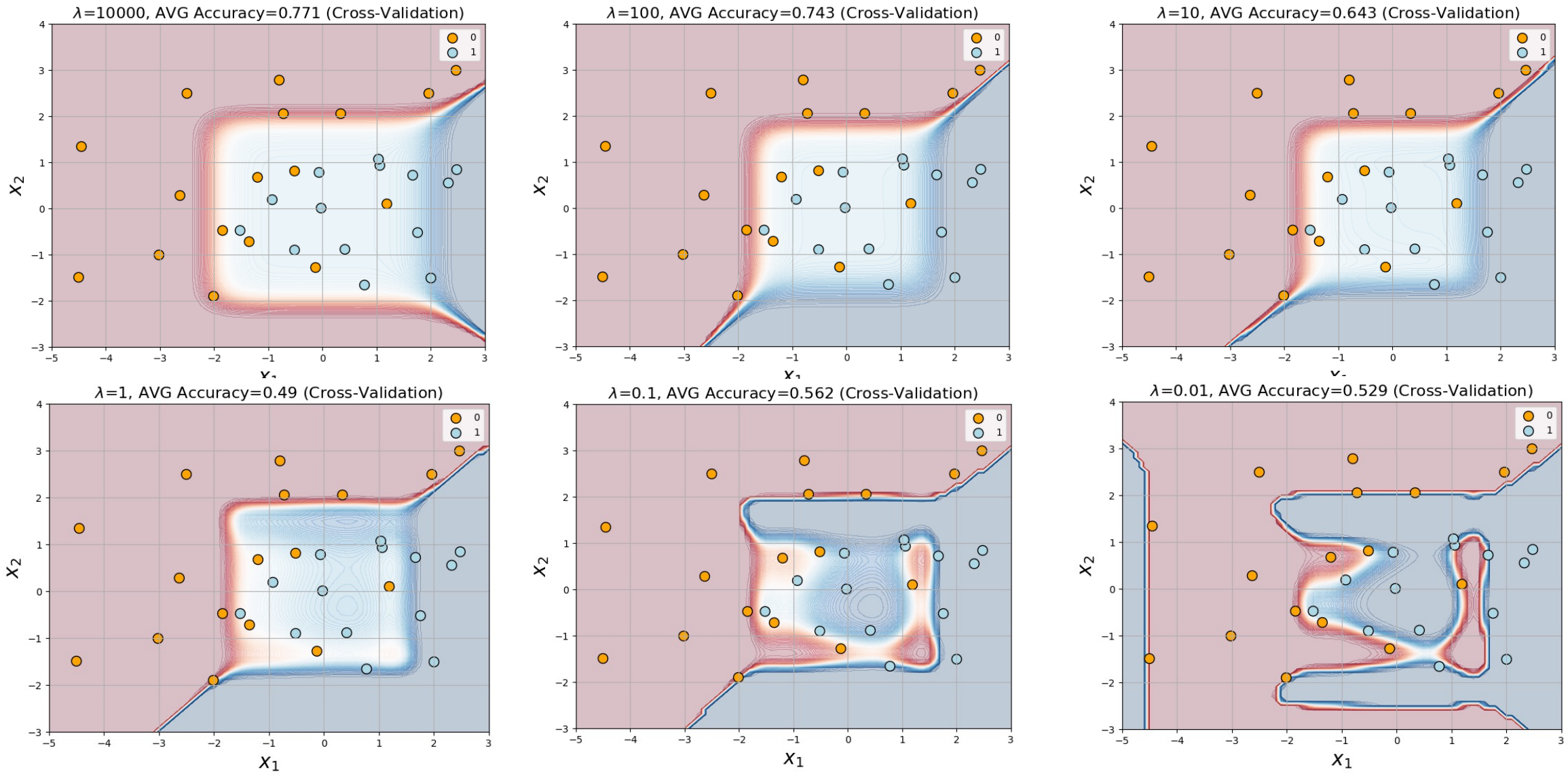
Regularised Decision Boundary (8th Order)

Feature	Coefficient
1	0.609
x_1	3.661e-4
x_2	-3.319e-5
x_1^2	-6.429e-5
x_2^2	-2.815e-5
x_1^3	8.919e-4
x_2^3	-1.843e-4
x_1^4	-1.316e-4
x_2^4	-2.531e-4
x_1^5	2.826e-3
x_2^5	-8.357e-4

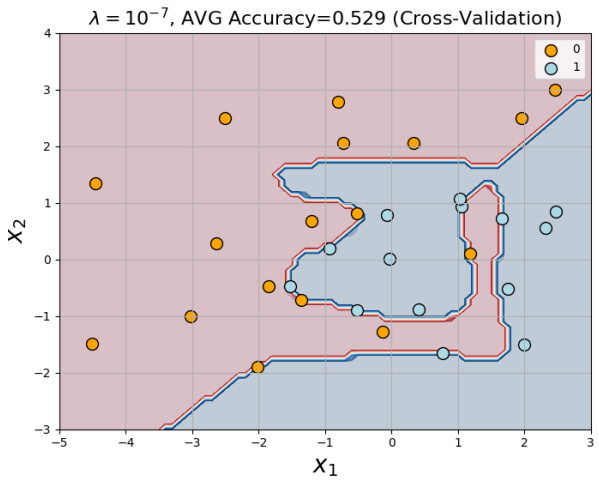
Feature	Coefficient
x_1^6	-2.408e-4
x_2^6	-1.386e-3
x_1^7	9.870e-3
x_2^7	-3.375e-3
x_1^8	-3.631e-4
x_2^8	-6.624e-3



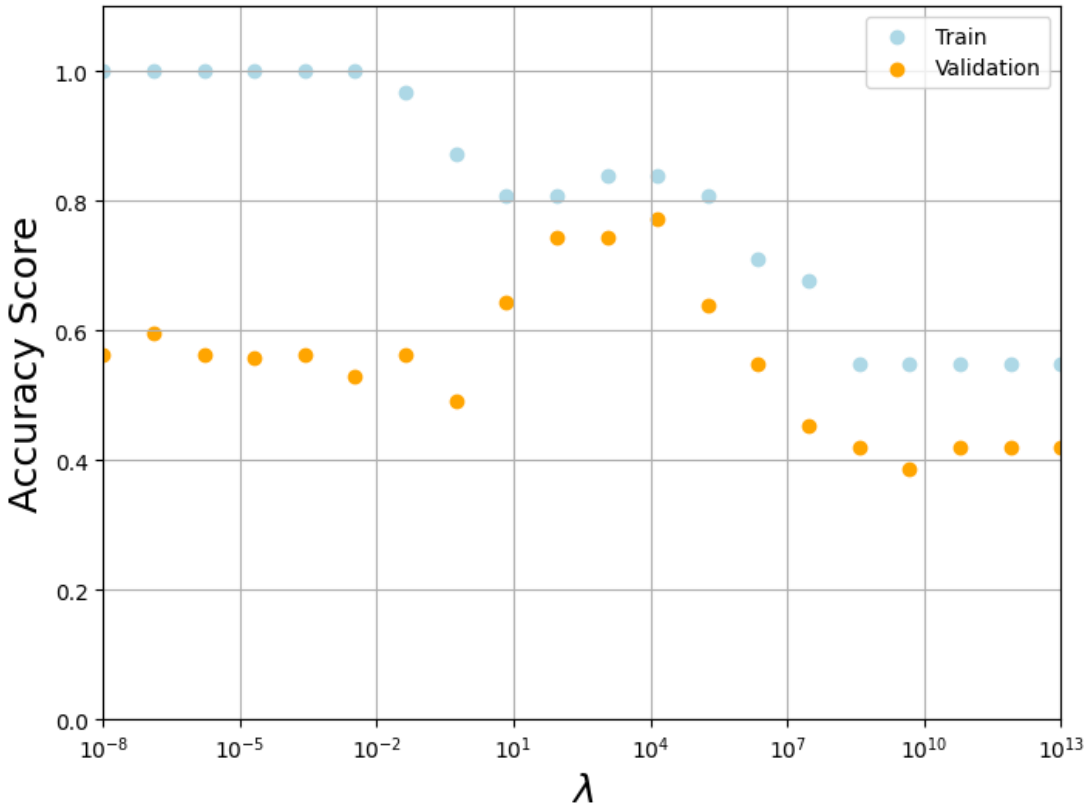
Model Complexity

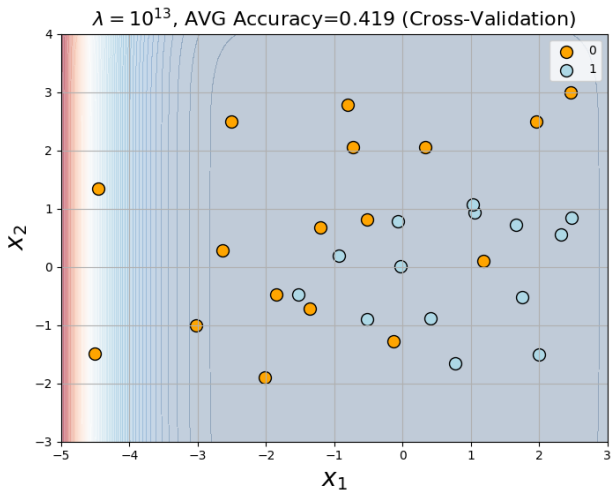


Bias-Variance Trade-Offs



- High Variance, Low Bias
- Complex Fitted Line
- Low Train Accuracy
- High Validation Accuracy



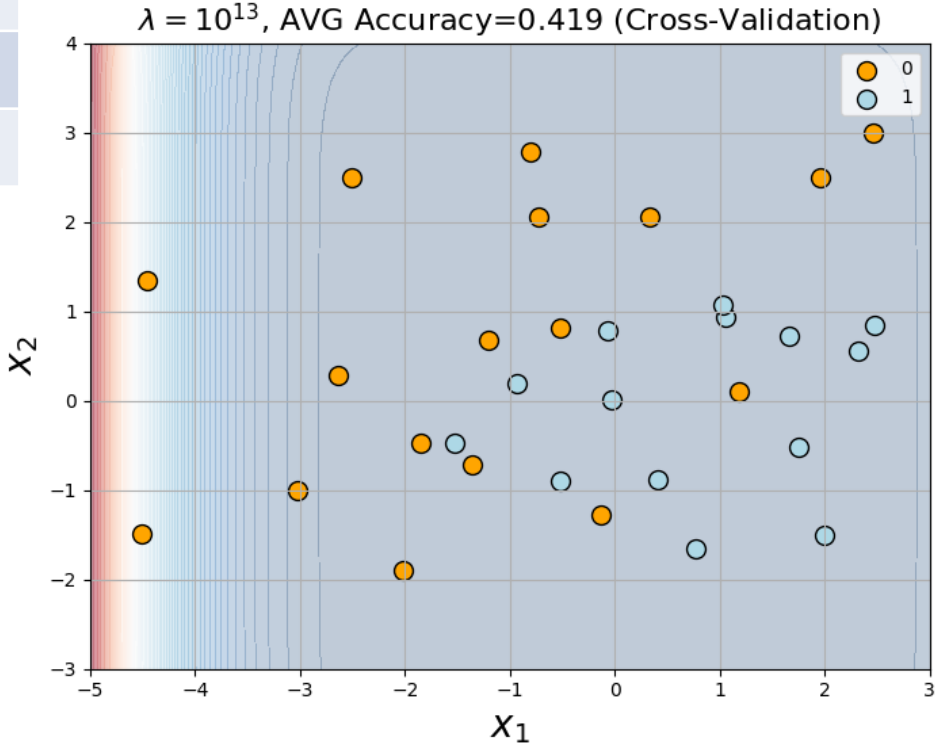


- High Bias, Low Variance
- Simple Fitted Line
- Low Train Accuracy
- Low Validation Accuracy

Regularised Decision Boundary ($\lambda = 10^{13}$)

Feature	Coefficient
1	-0.194
x_1	1.463e-12
x_2	-5.507e-13
x_1^2	-2.369e-12
x_2^2	-1.633e-12
x_1^3	1.310e-11
x_2^3	-4.208e-12
x_1^4	-4.208e-11
x_2^4	-1.171e-11
x_1^5	1.935e-10
x_2^5	-3.018e-11

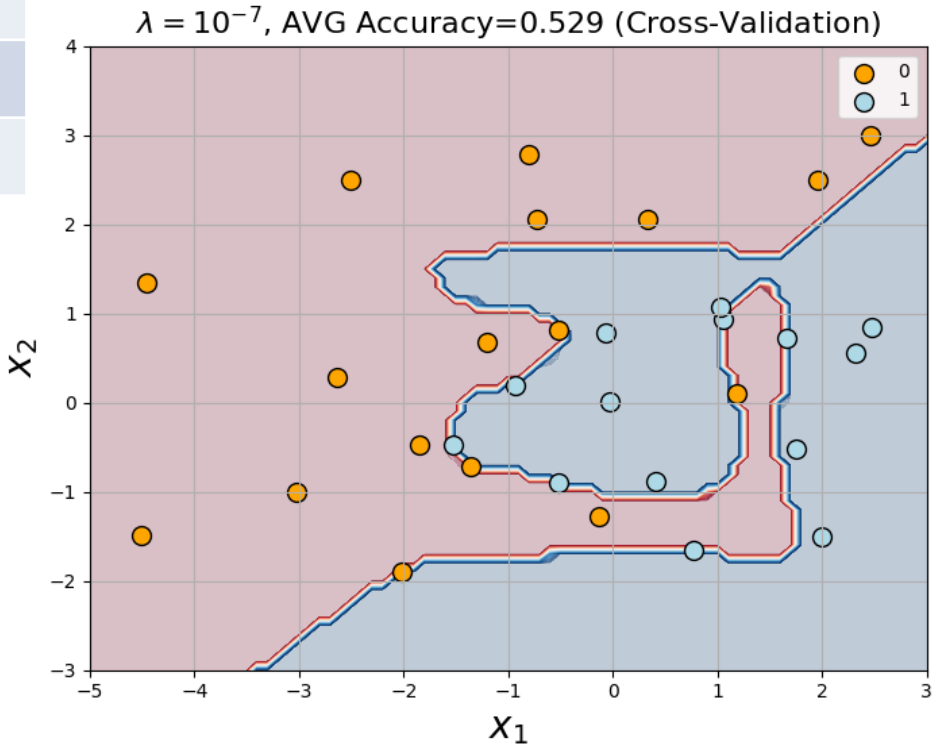
Feature	Coefficient
x_1^6	-7.800e-10
x_2^6	-8.390e-11
x_1^7	3.483e-9
x_2^7	-2.244e-10
x_1^8	1.508e-8
x_2^8	-6.308e-10



Regularised Decision Boundary ($\lambda = 10^{-7}$)

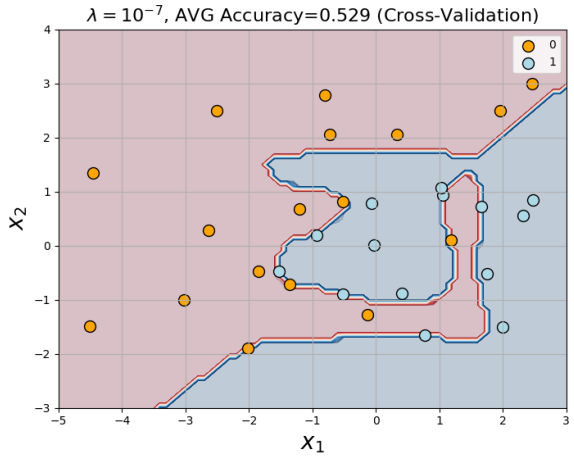
Feature	Coefficient
1	432.548
x_1	333.613
x_2	-352.095
x_1^2	-61.796
x_2^2	-256.131
x_1^3	-36.690
x_2^3	219.2734
x_1^4	-246.131
x_2^4	-106.567
x_1^5	-310.860
x_2^5	405.405

Feature	Coefficient
x_1^6	44.591
x_2^6	107.260
x_1^7	110.216
x_2^7	-150.206
x_1^8	11.905
x_2^8	15.687

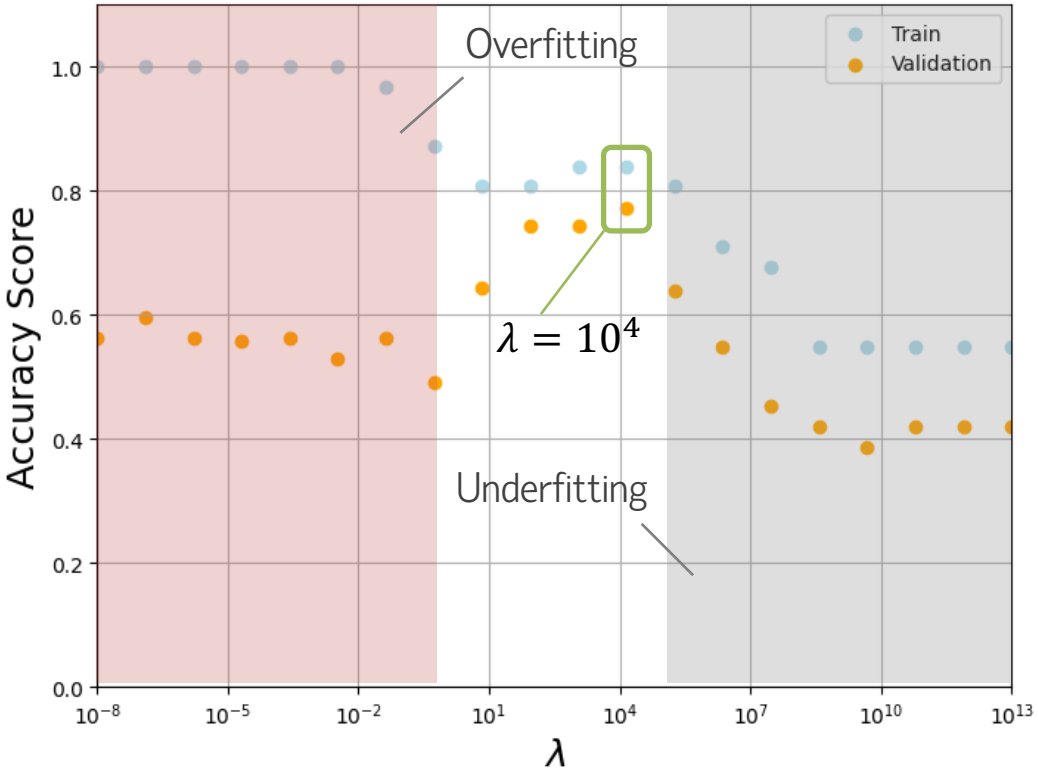


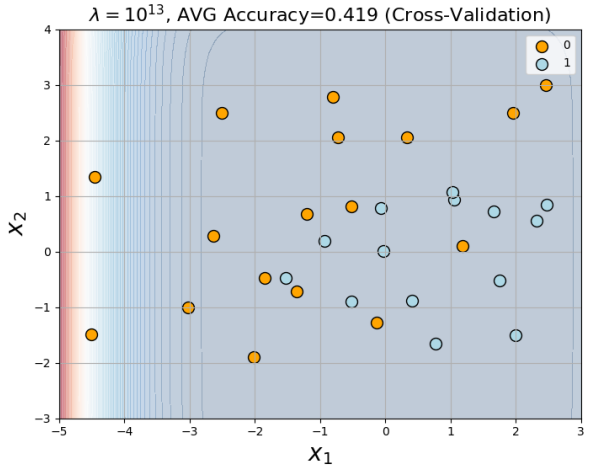
Overfitting vs Underfitting

- **Overfitting:** With minimal penalty on weights, the model captures noise in the training data, leading to high variance and poor generalization on validation data.
- **Underfitting:** A strong penalty on weights forces them to shrink, overly simplifying the model. This results in high bias, causing it to miss important patterns in both training and validation data.



- High Variance, Low Bias
- Complex Fitted Line
- Low Train Accuracy
- High Validation Accuracy





- High Bias, Low Variance
- Simple Fitted Line
- Low Train Accuracy
- Low Validation Accuracy

- **Optimal:** The ideal λ imposes just enough weight penalty to prevent overfitting while preserving the model's ability to capture true patterns, leading to accurate predictions on both training and unseen data.

MyLogisticRegressor (Revised)

```
import numpy as np
from scipy.special import expit  # Sigmoid function
from sklearn.base import BaseEstimator, ClassifierMixin

class MyLogisticRegressor(BaseEstimator, ClassifierMixin):
    def __init__(self, learning_rate=0.01, max_iter=1000, tol=1e-6, penalty='l2', lambda_=0.1):
        self.learning_rate = learning_rate
        self.max_iter = max_iter
        self.tol = tol
        self.penalty = penalty
        self.lambda_ = lambda_
        self.weights_ = None
```

MyLogisticRegressor (cont.)

```
    def _add_bias(self, X):
        """Add a column of ones to X to account for the bias term."""
        return np.hstack([np.ones((X.shape[0], 1)), X])

    def _compute_gradient(self, X, y, weights):
        """Compute the gradient of the cost function."""
        predictions = expit(X @ weights)  # Predicted probabilities using sigmoid function
        errors = y - predictions           # Error between actual and predicted
        gradients = X.T @ errors / X.shape[0]  # Average gradient

        # Add L2 regularization if penalty is set to 'l2'
        if self.penalty == 'l2':
            gradients -= self.lambda_ * weights

        return gradients

    def _gradient_descent(self, X, y):
        """Perform gradient descent to optimize weights."""
        weights = np.zeros((X.shape[1], 1))

        for iteration in range(self.max_iter):
            gradients = self._compute_gradient(X, y, weights)
            weights += self.learning_rate * gradients

            # Check for convergence
            if np.linalg.norm(gradients) < self.tol:
                print(f"Converged after {iteration} iterations.")
                break

        return weights
```

MyLogisticRegressor (cont.)

```
    def fit(self, X, y):
        """Fit the logistic regression model to the training data."""
        X = self._add_bias(X)
        y = y.reshape(-1, 1)  # Ensure y is a column vector
        self.weights_ = self._gradient_descent(X, y)
        return self

    def predict_proba(self, X):
        """Predict probability estimates for samples in X."""
        X = self._add_bias(X)
        probabilities = expit(X @ self.weights_)
        return np.hstack([1 - probabilities, probabilities])  # Return probabilities for classes [0, 1]

    def predict(self, X):
        """Predict class labels for samples in X."""
        proba = self.predict_proba(X)
        return (proba[:, 1] ≥ 0.5).astype(int)
```

Usage Example

```
# Example usage
if __name__ == "__main__":
# Generate some example data
np.random.seed(0)
m, n = 100, 2
X_class0 = np.random.randn(m, n) - 1
X_class1 = np.random.randn(m, n) + 1
X = np.vstack([X_class0, X_class1])
y = np.hstack([np.zeros(m), np.ones(m)])

# Initialize and fit MyLogisticRegressor
model = MyLogisticRegressor(learning_rate=0.1, max_iter=1000, tol=1e-6, penalty='l2', lambda_=0.1)
model.fit(X, y)

# Predict class probabilities and labels
print("Predicted probabilities:\n", model.predict_proba(X)[:5])  # Show probabilities for first 5 samples
print("Predicted labels:\n", model.predict(X)[:5])  # Show labels for first 5 samples
```

```
Converged after 426 iterations.

Predicted probabilities:
[[0.45729389 0.54270611] [0.25744809 0.74255191] [0.7140739 0.2859261 ] [0.73288984 0.26711016] [0.8095836 0.1904164 ]]

Predicted labels:
[1 1 0 0 0]
```

Summary

- Regularization helps prevent overfitting by penalizing large weights, encouraging models to capture meaningful patterns rather than noise. This leads to simpler, more generalized models that perform well on new data.
- The cost function in regularized logistic regression combines the negative log-likelihood with an added penalty term, $\lambda\|\vec{w}\|_2^2$. This term balances minimizing error with controlling weight magnitudes to reduce model complexity.
- The regularization parameter, λ, governs the strength of the penalty. A smaller λ may result in high variance and overfitting, while a larger λ increases bias and risks underfitting. Tuning λ helps strike an optimal balance between variance and bias.
- Regularization implicitly controls model complexity and performs feature selection by suppressing coefficients. Features with low-magnitude weights are essentially excluded from the model, allowing it to focus on the most impactful features and reducing overfitting.